Question Paper

Exam Date & Time: 30-Dec-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: MAT 2101/MAT_2101 DISCRETE MATHEMATICAL STRUCTURES
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

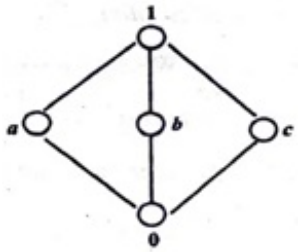
Answer all the questions.

1A) Let S = {a, b, c} and P(S) be the power set of S. Then draw the Hasse diagram of the poset (P(S), $\subseteq$). Is it a lattice? Justify your answer. (4)

Also list all the antichains of length more than one in the poset P(S).

Course Outcome: "CO1" / Cognitive Level: "Analysis"

1B) i. Verify whether the following lattice is distributive. Justify your answer. (3)



ii. If a lattice (A, $\leq$) is distributive then show that for any elements a, b, c in A, $(a \vee b) \wedge c \leq a \vee (b \wedge c)$

Course Outcome: "CO1" / Cognitive Level: "Analysis"

1C) Let $E(x_1, x_2, x_3) = \overline{x_1} \vee \left(\overline{x_2} \vee (\overline{x_3} \wedge x_1)\right)$ be a Boolean expression over the two-valued Boolean algebra $(\{0,1\}, \vee, \wedge, \bar{})$. Write $E(x_1, x_2, x_3)$ in both disjunctive and conjunctive normal forms. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

2A) Determine the number of integers between 1 and 250 (including 1 and 250) that are not divisible by any of the integers 2,3,5 and 7. (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2B) Show that the number of partitions of n is equal to the number of partitions of 2n which have exactly (3)

n parts.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2C) Find the 60th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

3A) (4)

i) Prove that a nonempty subset H of a group (G, *) is a subgroup of G if and only if $a * b^{-1} \in H$ for every $a, b \in H$

ii) Define $*$ on $R - \{0\}$ as $a * b = \frac{ab}{5}$, for all $a, b \in R - \{0\}$, where R is a set of all real numbers. Show that $(R - \{0\}, *)$ forms an abelian group.

Course Outcome: "CO4" / Cognitive Level: "Application"

3B) If G is any graph with 6 vertices then show that either G or $\bar{G}$ contains a triangle. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

3C) Solve $2x_1 + x_2 - x_3 = 1$
$x_1 + 2x_2 + x_3 = 8$
$-x_1 + x_2 - x_3 = -5$ by LU decomposition method. (3)

Course Outcome: "CO6" / Cognitive Level: "Application"

4A) (4)

i. If G is a finite group and H a subgroup of G, then show that the order of H divides the order G.

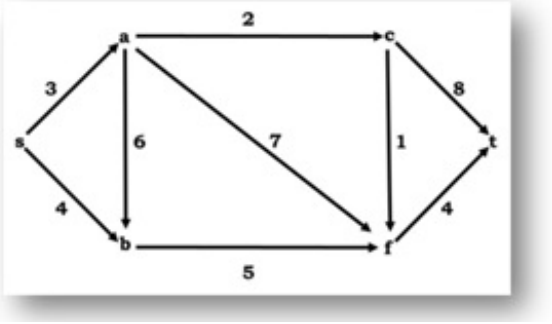
ii. Is the group {1,2,3,4,5,6} under the operation "multiplication modulo 7" cyclic? Justify your answer.

Course Outcome: "CO4" / Cognitive Level: "Application"

4B) In a tree with 14 pendant vertices, the degree of every non-pendant vertex is either 4 or 5. Find the number of vertices of degree 4 and 5, respectively. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

4C) Using Dijkstra's algorithm, find the shortest paths from 's' to all the other vertices. (3)



Course Outcome: "CO3" / Cognitive Level: "Application"

5A) Find a singular value decomposition of the matrix (4)

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Course Outcome: "CO6" / Cognitive Level: "Application"

5B) Show that s ∨ r is tautologically implied by the premises p ∨ q, p → r, and q → s. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5C) Prove that $(\exists x)(D(x) \wedge A(x)),\ (\forall x)(A(x) \rightarrow C(x)) \Rightarrow (\exists x)(D(x) \wedge C(x))$. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
MAKE-UP EXAM-ODD SEMESTER 2025 MIT MANIPAL	MAT 2101:Discrete Mathematical Structures

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	9
4	4	CO 4	8
5	5	CO 5	6
6	6	CO 6	7
		Total	**50**

* NA - CO Not Selected

Event Name	Subject
MAKE-UP EXAM-ODD SEMESTER 2025 MIT MANIPAL	MAT 2101:Discrete Mathematical Structures

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	33(66%)
4	Analysis	5	17(34%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)